Rethinking case attraction on Ancient Greek infinitive clauses

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(1) a. συμβουλεύει **τῷ Ξενοφῶντι ἐλθόντα** εἰς Δελφοὺς ἀνακοινῶσαι τῷ θεῷ περὶ τῆς πορείας.
He advices Xenophon to go to Delphi and ask the god about the travel. (Xen. *Anab.* 3.1.5)

b. ἀφῆκε **μοι ἐλθόντι** πρὸς ὑμᾶς λέγειν τἀληθῆ.
He allowed me to go and speak the truth in front of you. (Xen. *Hell.* 6.1.13)

Mentioned works: Lancelot (1658, 1709), Boas et al. (2019), Lakoff (1970), Andrews (1971), Quicoli (1982), Spyropoulos (2005), Tantalou (2003), Sevdali (2007, 2013)

Data source: Diorisis Corpus (Vatri and McGillivray 2018, 2020) with help of the project's querying tool, Diorisis Search (Vatri 2020-2022).

Corpus: Sentences collected from the attic literary sources and Herodotus.

Tooling: R (R Core Team 2013), the packages tidyverse (Wickham et al. 2019), dagitty (Textor et al. 2016), ggdag (Barrett 2025), and rethinking (McElreath 2020), as well as the software Stan (Stan Development Team 2025a,b).

Typological framework: Corbett (2006)

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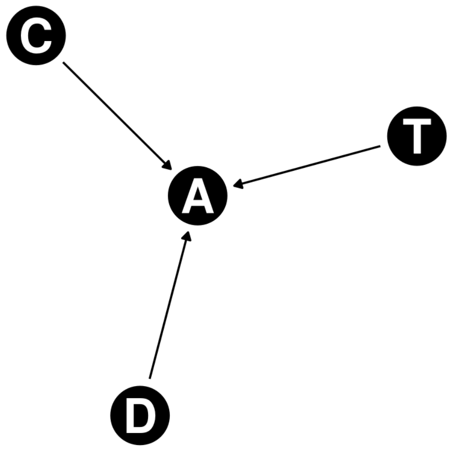


Copular infinitives: proposed in by Kühner and Gerth (1898)

(2) Agreement and Predicate Hierarchies (Corbett 2006, p. 233):

Attributive → Predicative Verb → Participle → Adjective → Noun

Predicative Verb → Relative Pronoun → Personal Pronoun

\- ⟷ +

Likelihood of agreement with semantic justification

Impersonal verbs: see Humbert (1945)

Verbs with modal semantics: verbs such as ἐξαρκεῖ and ἔξεστί have higher probability to display case attraction (Geraldes 2020, 2023).

(3) ἀλλ’ ἐξαρκῇ σοι τὰ ἔργα αὐτῶν ὁρῶντι σέβεσθαι καὶ τιμᾶν τοὺς θεούς.
But it is enough for you to look at these deeds and respect the gods. (Xen. *Mem.* 4.3.13)

Authorship and dialect: Geraldes (2020, 2023) shows that authors use verbs of modal semantic more often in philosophical dialogues.

Word order and Distance: longer distances diminish the likelihood of case attraction. Targets preceding the controller seem to seem to demand case attraction in examples such as (4).

(4) ὡς **ἀγαθοῖς** τε **ὑμῖν** προσήκει εἶναι.
That you may be good people. (Xen. *Anab.* 3.2.11)

Word order and communicative function: Word Order is not 1-1 equivalent to focus, topic and any other communicative function. It makes an element more likely to be more topical or a focus, but there are unobserved features.

Distance flows from Word Order: Since distance depends on word order, we can only estimate the direct effect by controlling for Word Order.

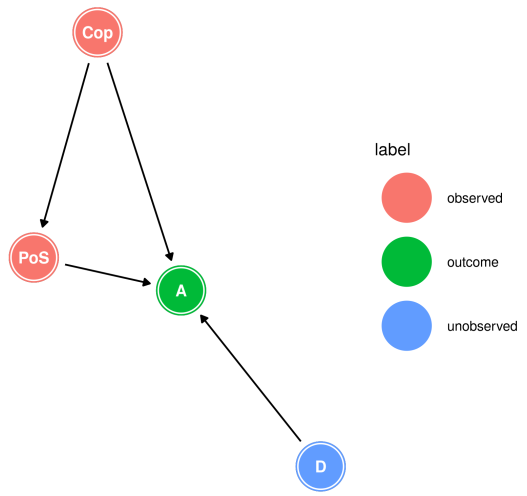


Figure 1: DAG with **A**ttraction, **D**omain, **P**art **o**f **S**preech and **Cop**ula

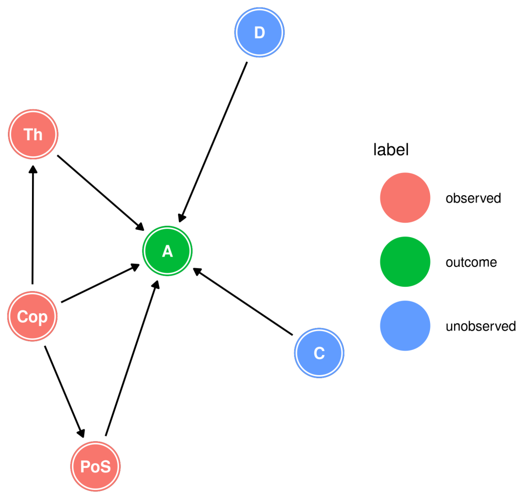


Figure 2: DAG with **A**ttraction, **D**omain, **C**ontroller **P**art **of** **S**preech, **Cop**ula and **Th**ematic role.

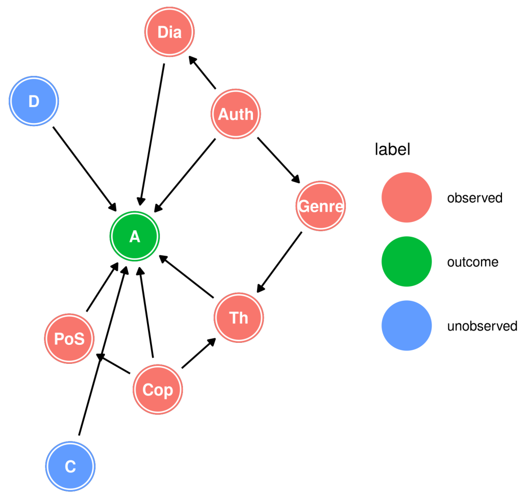


Figure 3: DAG with **A**ttraction, **D**omain, **C**ontroller **P**art **of** **S**preech, **Cop**ula, **Th**ematic role, **Auth**orship, **Genre**, and **Dia**lect.

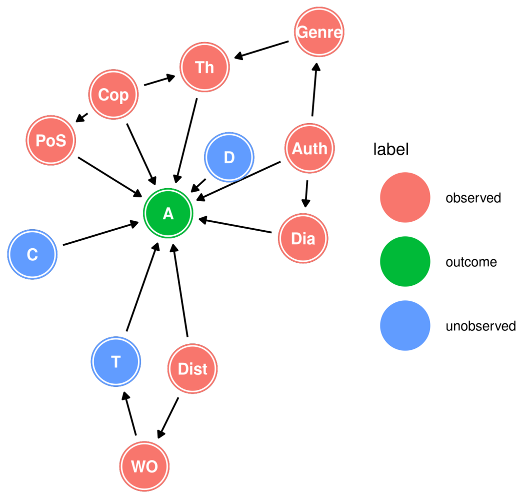
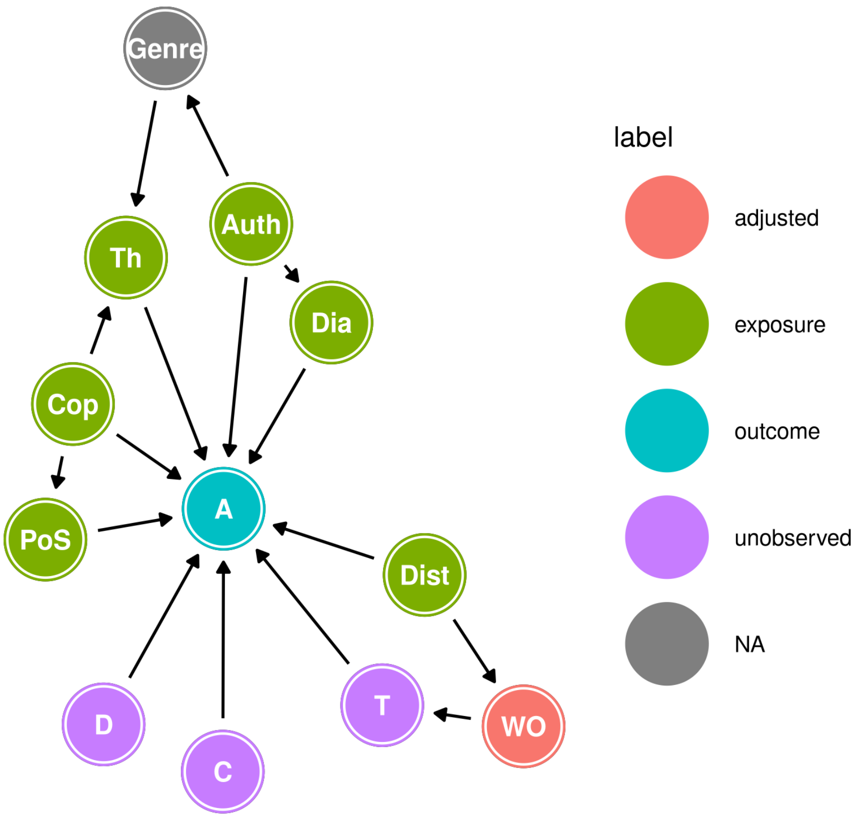


Figure 4: DAG with **A**ttraction, **D**omain, **C**ontroller, **T**arget, **P**art **of** **S**preech, **Cop**ula, **Th**ematic role, **Auth**orship, **Genre**, **Dia**lect, **Dist**ance, and **W**ord **O**rder.

Final causal model:



Adjustment set: WO

Generalized linear model:

$$
\begin{aligned}
\text{A} &\sim \text{Bernoulli}(p_i) \\
\text{logit}(p_i) &= \alpha + \beta_{\text{Cop}}\text{Cop}_i + \beta_{\text{Dist}}\text{Dist}_i + \beta_{\text{WO}}\text{Prec}_i \\
&\quad + \beta_{\text{PoS}} \sum_{j=0}^{PoS_i - 1} \delta_{j_{\text{PoS}}} + \beta_{\text{Th}} \sum_{j=0}^{Th_i - 1} \delta_{j_{\text{Th}}} \\
&\quad + \beta_{\text{Auth}_i} + \beta_{\text{Dia}_i} \\
\alpha &\sim \text{Normal}(0, 1) \\
\beta_{\text{Cop}}, \beta_{\text{Dist}}, \beta_{\text{Prec}} &\sim \text{Normal}(0, 1) \\
\beta_{\text{PoS}}, \beta_{\text{Th}} &\sim \text{Normal}(0, 0.5) \\
\delta_{\text{PoS}}, \delta_{\text{Th}} &\sim \text{Dirichlet}(2) \\
\beta_{\text{Auth}} &= z_{\text{Auth}} * \sigma_{\text{Auth}} \\
\beta_{\text{Dia}} &= z_{\text{Dia}} * \sigma_{\text{Dia}} \\
z_{\text{Auth}}, z_{\text{Dia}} &\sim \text{Normal}(0, 1) \\
\sigma_{\text{Auth}}, \sigma_{\text{Dia}} &\sim \text{Exponential}(1)
\end{aligned}
$$

Copula and Distance

Effect	mean	sd	5.5%	94.5%	rhat
β_{Copula}	1.60	0.37	1.02	2.20	1
β_{Dist}	-0.21	0.04	-0.28	-0.15	1

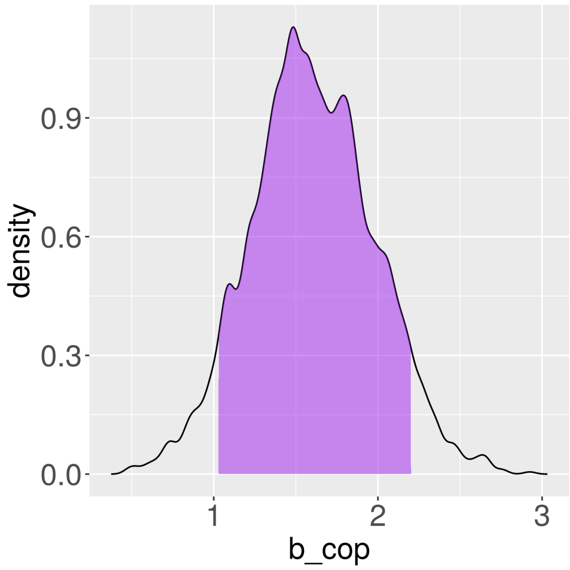


Figure 5: Posterior distribution of β_{Cop} with the 89% Highest Posterior Density Interval.

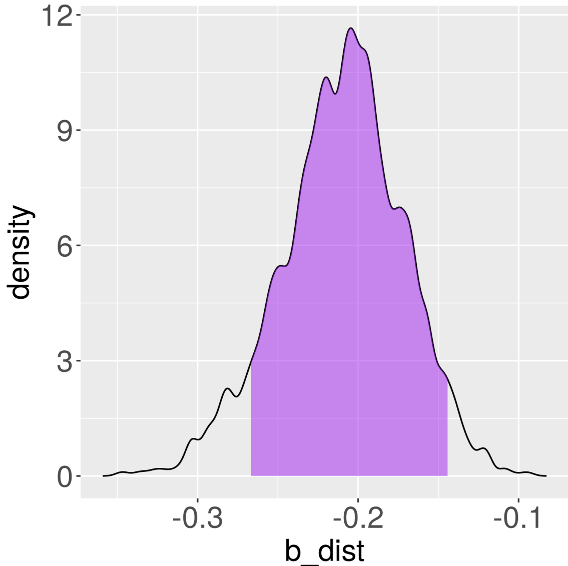
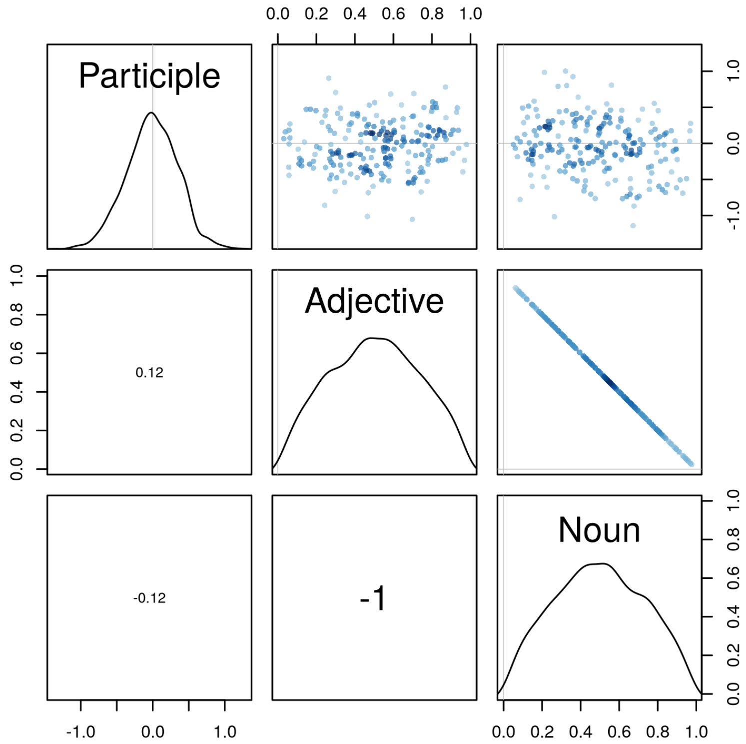


Figure 6: Posterior distribution of β_{Dist} with the 89% Highest Posterior Density Interval.

Part of Speech and the Predicate Hierarchy

Effect	mean	sd	5.5%	94.5%	rhat
β_{PoS}	-0.01	0.35	-0.59	0.52	1
δ_{Adj}	0.50	0.23	0.12	0.88	1
δ_{Noun}	0.50	0.23	0.12	0.88	1

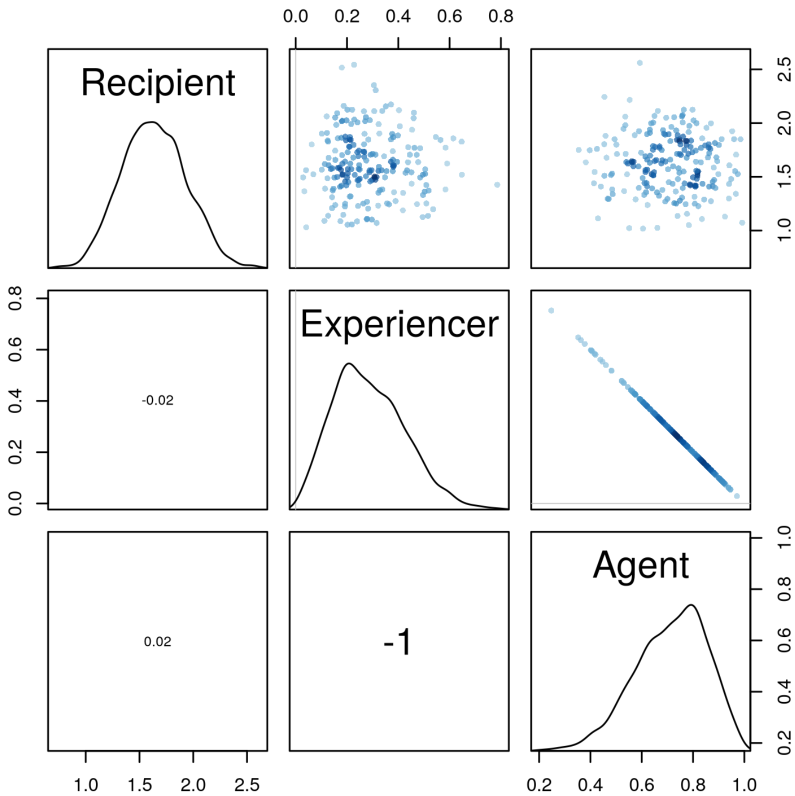
PoS	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Participle	β_{PoS}	-0.01	0.35	-0.59	0.52	1
Adjective	$\beta_{\text{PoS}} + \delta_{\text{Adj}}$	0.49	0.45	-0.22	1.19	1
Noun	$\beta_{\text{PoS}} + \delta_{\text{Adj}} + \delta_{\text{Noun}}$	0.99	0.35	0.41	1.52	1



Thematic Role of the Controller

Effect	mean	sd	5.5%	94.5%	rhat
β_{Theta}	1.64	0.30	1.17	2.12	1
δ_{Exp}	0.29	0.14	0.09	0.52	1
δ_{Agent}	0.71	0.14	0.48	0.91	1

Theta	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Recipient	β_{Rec}	1.64	0.30	1.17	2.12	1
Experiencer	$\beta_{\text{Rec}} + \delta_{\text{Exp}}$	1.92	0.33	1.41	2.44	1
Agent	$\beta_{\text{Rec}} + \delta_{\text{Exp}} + \delta_{\text{Agent}}$	2.64	0.30	2.17	3.12	1



Dialect and Authorship

Effect	mean	sd	5.5%	94.5%	rhat
$\overline{\beta}_{\text{Dia}}$	-0.22	0.86	-1.58	1.15	1
σ_{Dia}	1.36	0.78	0.50	2.78	1
β_{Attic}	0.45	0.87	-0.93	1.80	1
β_{Ionic}	-1.35	0.97	-2.94	0.18	1
$\beta_{\text{Attic}} - \beta_{\text{Ionic}}$	1.80	0.63	0.78	2.78	NA

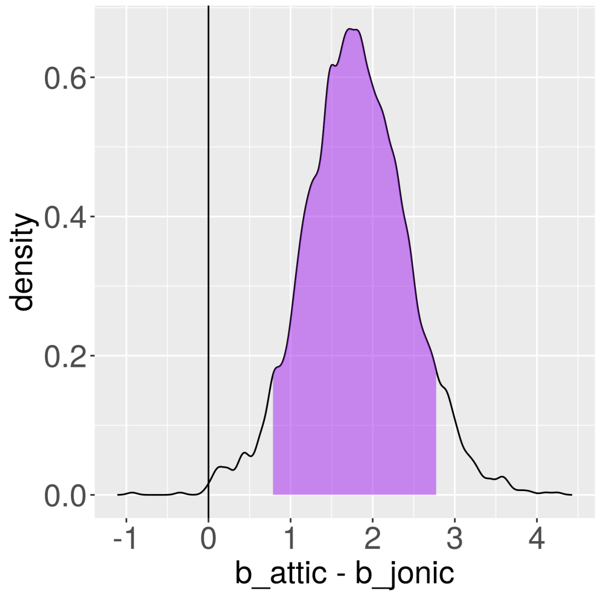
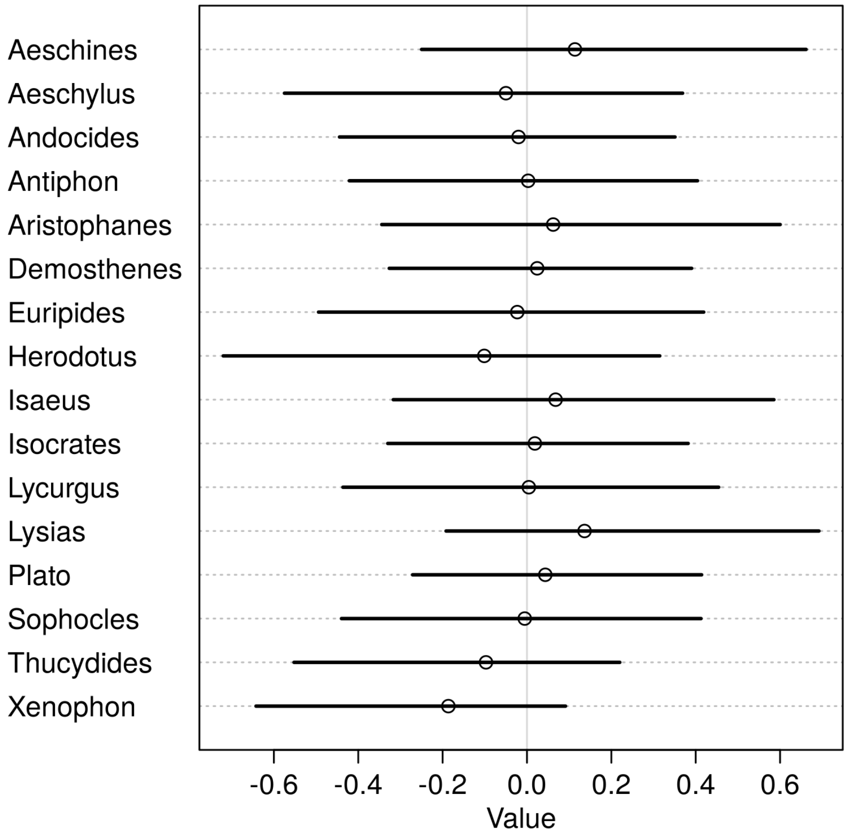


Figure 7: Posterior distribution of the difference between β_{Attic} and β_{Jonic} with the 89% Highest Posterior Density Interval.



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